

Translating KDE Software Into Taiwanese Taigi

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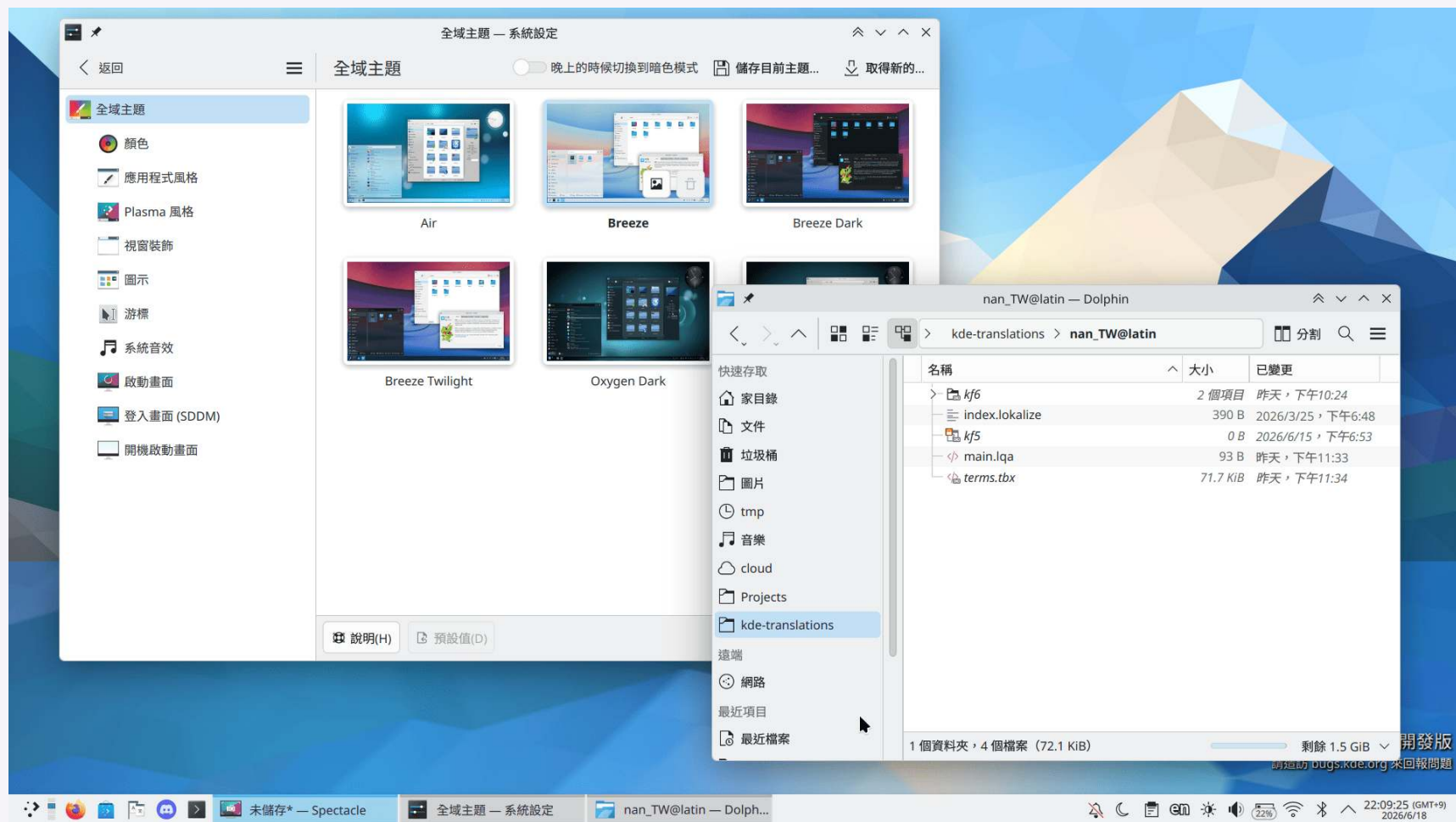
Introduction

Who am I

- Shen Yu-hsiang / Kisaragi Hiu
- Main work: 2022~ on zh_TW KDE translations

What: KDE

- KDE Plasma: major Linux desktop UI (other is GNOME)
 - Steam Deck desktop mode
- KDE: a free software community w/ common infra & tech stack (Qt)
- Other KDE apps: Dolphin, Krita, Calligra, Okular
- Unified translation infrastructure



KDE Plasma

What: Taiwanese Taigi

- aka Taiwanese
- ^ Could confuse as canonical language of Taiwan
- aka Taiwanese Hokkien
- ^ Many prefer its identity recognized
- → National Language Development Report (MOC, 2022) compromise: emphasize Taigi is not the only language of Taiwan

Why

- No Taigi user interface translations available
- Provide options for those who prefer
- I already contribute zh_TW translations to KDE, probably able to help

What work

- Start translating KDE software into Taiwanese Taigi
- Infrastructure: modern tech terms
- Establish a language team
- Prerequisite before submitting upstream:
kcoreaddons, kio, kxmlgui
 - Filetypes, licenses, some units; file IO and file management; common UI library

Taigi software prior work

- Pleroma translated by Tân Kiàn-ting in 2015¹
- Mastodon also translated by Tân Kiàn-ting in 2025²
- Southern Min Wikipedia³

¹陳建町. (2015). 參與自由軟體台文化 ê 一寡心得. https://tsbp.tgb.org.tw/2015/05/e_92.html

²陳建町. (2025). *Add `nan\_TW` to interface languages - mastodon/mastodon*. <https://github.com/mastodon/mastodon/pull/34923>

³Wikipedia contributors. (n.d.). *Southern Min Wikipedia*. Retrieved <https://zh-min-nan.wikipedia.org/>

Foundational choices

Language name

- National Language Development Report:
“Taiwanese Taigi” or “Taigi”
- ISO 639-3 etc.: “Min Nan Chinese”
- Can’t fight that apart from CR 2021-044 (still pending)
- → “Taiwanese Taigi” / “Taigi” if possible,
otherwise “Min Nan Chinese”

aside: ISO 639-3 Change Request 2021-044

- Split Taigi out from Min Nan Chinese (nan) and give it ftg
- Would officially call Taigi “Taigi”
- Submitted by Ìthuan Khoki (意傳科技) in 2021
- Still pending to this day
- Fingers crossed?

Language code

- → nan_TW for now
- Hopefully CR 2021-044 will get accepted?
If so we'd get to use ftg

Writing system

- “No standard writing system”?

Writing system

- Han characters, romanization (Tâi-lô or POJ), Kana, Bopomofo, Hangul...
- Too many

Writing system

- Han characters, romanization (Tâi-lô or POJ),
~~Kana~~, Bopomofo, Hangul...
 - Unused since 1945

Writing system

- Han characters, romanization (Tâi-lô or POJ),
Kana, Bopomofo, Hangul...
 - Transcription scheme, not used as primary orthography

Writing system

- Han characters, romanization (Tâi-lô or POJ),
~~Kana, Bopomofo, Hangeul...~~
 - Not seen it ever (no idea why English Wikipedia lists it)

Writing system: POJ or TL

- Tâi-lô is an OK spelling reform
(ch → ts; oe/oa → ue/ua; eng → ing; tones sometimes on different vowels)
- Perfectly usable as primary orthography
- Clear standard
- No conflict (p is always p, 1-to-1 differences)
- → Use Tâi-lô

Writing system

- Han characters, romanization (Tâi-lô or ~~P~~ΘJ), Kana, Bopomofo, Hangeul...
- Really just 3 valid standard options
- Standards: MOE dictionary, POJ Bible and conventions, 57-page Tâi-lô definitive manual⁴

⁴Ministry of Education. (2024). 臺灣台語羅馬字拼音方案使用手冊. <https://language.moe.gov.tw/uploads/files/17579930721182.pdf>

Mixed writing

- Choose 2: like Serbian, Uzbek, Mandarin...
- all-Han: inconvenient
(even MOE dictionary has eg. oo-tóo-bái⁵)
- all romanized (tsuân-lô): decades of publishing
- mixed script (hàn-lô): decades of publishing
- → Both **mixed script** and **all romanized**

⁵陳建中. (2023). 教典的「漢羅濫寫」例句. <https://kiantiong.com/sutian/2/>

Glibc locale

- Locale: name, datetime format, numeric format, etc.
- nan_TW (all-Han or mixed) locale already available since 2013⁶
- nan_TW@latin (all-romanized) locale already available since 2009⁷

⁶Wei-Lun Chao. (2013). *New locale: nan_TW*. https://sourceware.org/bugzilla/show_bug.cgi?id=15608

⁷Arne Goetje. (2009). *New locale: nan_TW@latin*. http://sourceware.org/bugzilla/show_bug.cgi?id=9916

Translation process

KDE translation process

- GNU Gettext norms
- Source code → extract to template → translate → incorporate translation into source code
- KDE Invent repositories → KDE Subversion repository → Translate & commit → KDE Invent repositories, automated by Scripty

Creating a new language team

- Metadata, send email to kde-l10n-doc mailing list announcing it, discuss, translate kcoreaddons, kio, kxmlgui, send if accepted
- Compromise on language name was made here

Translating

- Using Lokalize (local CAT application, itself a KDE application)
- Set up workspace so strings are synced
- `nan_TW@latin` first to make sure using Taigi words, then write again in `nan_TW`

Project Overview

Glossary

kio6.po

Search

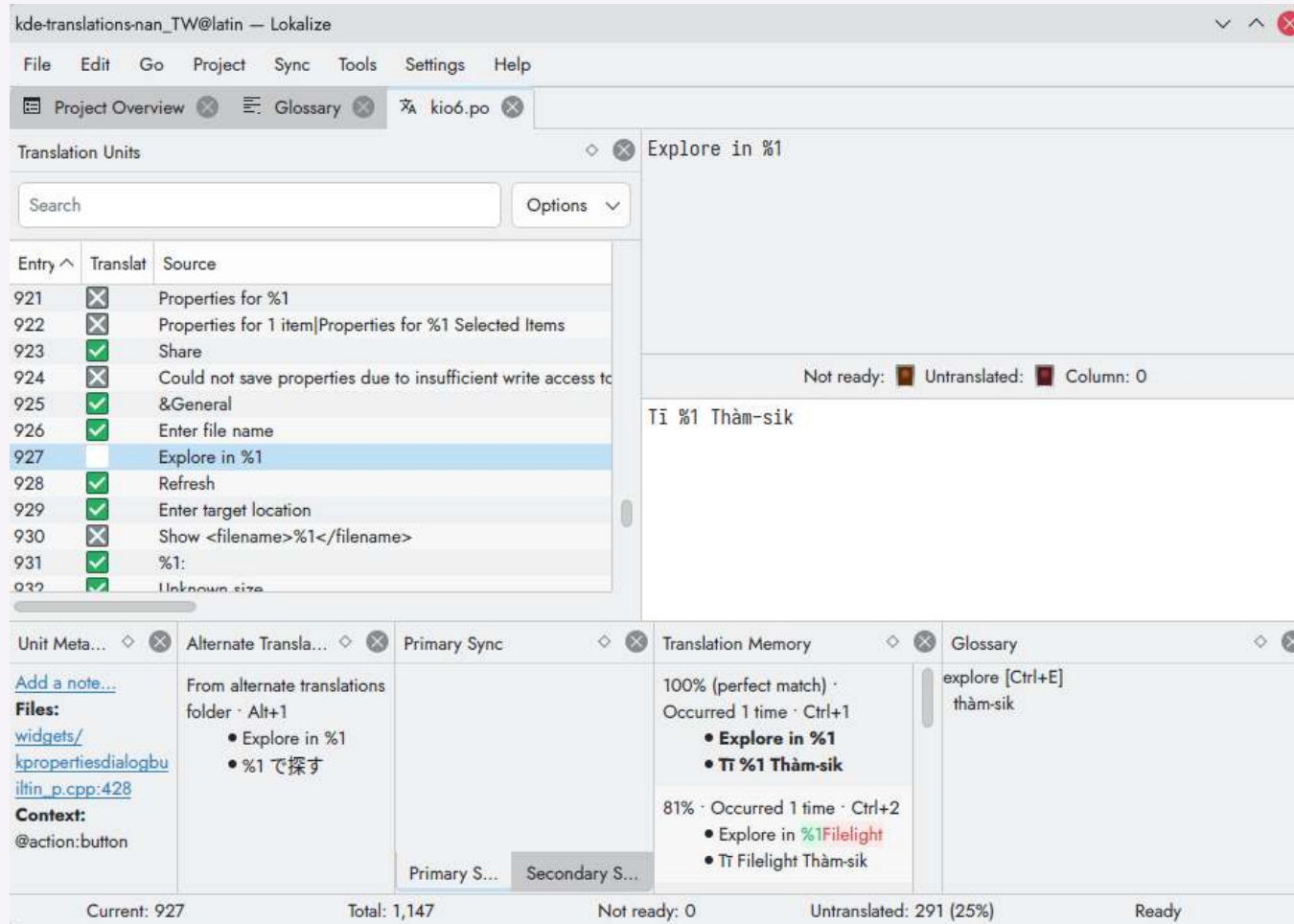
Name	Graph	Total	Translated	Incomplete	Untranslated
kf6		212,897	2,039	210,858	210,858
messages		212,897	2,039	210,858	210,858
kconfig		206	206	0	0
kconfigwidgets		223	223	0	0
kconfigwidgets._desktop_.po		106	106	0	0
kconfigwidgets6.po		117	117	0	0
kcoreaddons		217	217	0	0
kcoreaddons6_qt.po		162	162	0	0
kde6_xml_mimetypes.po		55	55	0	0
kxmlgui		201	201	0	0
kxmlgui6.po		201	201	0	0
kio		1,483	1,192	291	291
kio._desktop_.po		332	332	0	0
kio._json_.po		4	4	0	0
kio6.po		1,147	856	291	291
akonadi-import-wizard		94	0	94	94
akonadi-import-wizard._desktop_.po		2	0	2	2
akonadi-import-wizard._json_.pot		8	0	8	8
akonadiimportwizard.pot		84	0	84	84
akonadi-mime		117	0	117	117
akonadi-search		22	0	22	22
akonadiconsole		372	0	372	372
akregator		475	0	475	475
alnaka		57	0	57	57

Total: 329,395

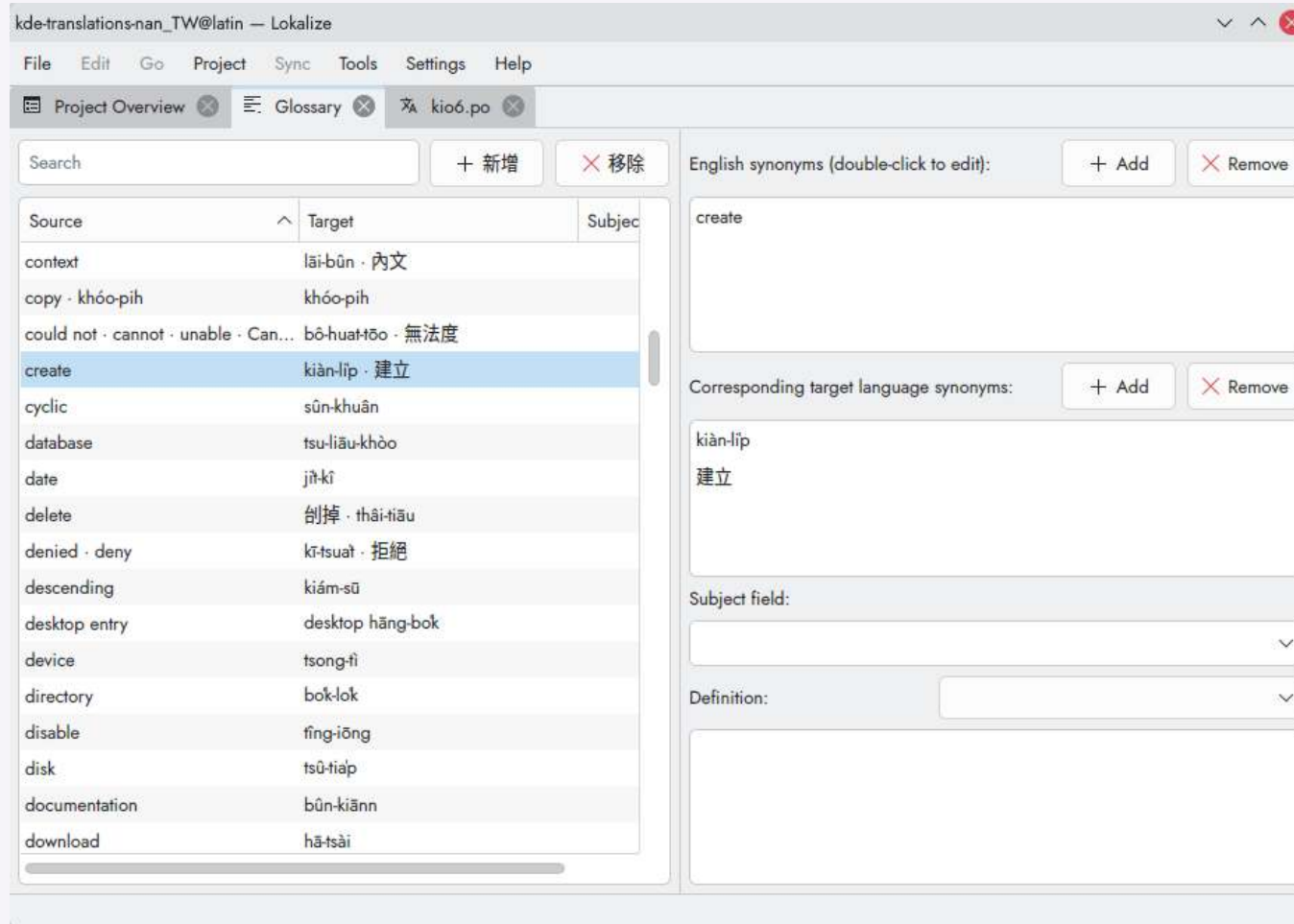
Not ready: 0

Untranslated: 327,356 (99%)

Project overview



Translating strings in a file



Glossary management

Word choices

- experience: 5% the essence, 95% the discourse
- Use existing if possible (dictionaries, PTS Taigi TV's Taigi Words, search in ChhoeTaigi or Kemdict)
 - Kemdict: my own dictionary search frontend, slightly different, need both it and ChhoeTaigi
- Borrow: directly from English, directly from another source language, orthographically from Mandarin...

Word choices: examples

- Email: 電子批信 (seen in ChhoeTaigi's interface)
- Catalanian: Catalunya-gí
- Desktop environment: Toh-bīn-khuân-kíng (桌面環境)
Calqued from Mandarin / also orthographic borrowing
- Open Containing Folder: 拍開保存資料鋏仔
Japanese 保存フォルダを開く
(lit. Open the folder where [the file] is saved)

Results

Results

- As of sending: base modules kcoreaddons, kio, kxmlgui + helpful kconfig, kconfigwidgets:
 - nan_TW@latin 79% words (10828/13639), 92% strings (2147/2330)
 - nan_TW 65% words (8975/13639), 83% strings (1947/2330)
- June 15, 2026: Team created! (Thank you aacid)
https://l10n.kde.org/team-infos.php?teamcode=nan_TW
https://l10n.kde.org/team-infos.php?teamcode=nan_TW@latin
- Staging repo:
https://github.com/kisaragi-hiu/kde-l10n-nan_tw

Phah-khui(O)...	Ctrl+O
Khui Kîn-lâi lōng-ê(R)	
Tû-tsûn(S)	Ctrl+S
lōng Pat-ê Miâ Tû-tsûn(A)...	Ctrl+Shift+S
Khóo-pih	Ctrl+C
Copy Text (Without line breaks)	Ctrl+Shift+C
Tshuē(F)...	Ctrl+F
Hok-guān	Ctrl+Z
Tshú-siau hok-guān	Ctrl+Shift+Z
Liān(P)...	Ctrl+P
Tlám Liān(W)	

Gwenview	
檔案(F)	編輯(E) 檢視(V) 前往(G) 設定(S)
拍開(O)...	Ctrl+O
開近來用-ê(R)	>
儲存(S)	Ctrl+S
用別个名儲存(A)...	Ctrl+Shift+S
Reload	F5
列印(P)...	Ctrl+P
預覽列印(W)	
Add Folder to Places	
Rename...	F2
Trash	Del
剷掉(D)	Shift+Del

nan_TW@latin in Okular, nan_TW translations in Gwenview

Discussion

Future work & challenges

- More translation work, challenging (esp. with lack of manpower) but feasible
- Website, better/more collaboration spaces, word choice overview web page
- Lokalize work (multiple reference languages, glossary bugfix)

Declarations

- zh_TW translator for the last 5 years, KDE e.V.
voting member for this
- AI use: occasional chatbot use, pretranslation for a list of languages, grammar review for the report

Bibliography

Arne Goetje. (2009). *New locale: nan_TW@latin*. http://sourceware.org/bugzilla/show_bug.cgi?id=9916

Ministry of Education. (2024). 臺灣台語羅馬字拼音方案使用手冊. <https://language.moe.gov.tw/uploads/files/17579930721182.pdf>

Wei-Lun Chao. (2013). *New locale: nan_TW*. https://sourceware.org/bugzilla/show_bug.cgi?id=15608

Wikipedia contributors. (n.d.). *Southern Min Wikipedia*. Retrieved <https://zh-min-nan.wikipedia.org/>

陳建中. (2023). 教典的「漢羅濫寫」例句. <https://kiantiong.com/sutian/2/>

陳建町. (2015). 參與自由軟體台文化 ê 一寡心得. https://tsbp.tgb.org.tw/2015/05/e_92.html

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